

Assessing the cost effectiveness of using prognostic biomarkers with decision models: case study in prioritising patients waiting for coronary artery surgery

SCIFACT-EVIDENCE-13625993

Evidence abstract

OBJECTIVE To determine the effectiveness and cost effectiveness of using information from circulating biomarkers to inform the prioritisation process of patients with stable angina awaiting coronary artery bypass graft surgery.

DESIGN Decision analytical model comparing four prioritisation strategies without biomarkers (no formal prioritisation, two urgency scores, and a risk score) and three strategies based on a risk score using biomarkers: a routinely assessed biomarker (estimated glomerular filtration rate), a novel biomarker (C reactive protein), or both.

The order in which to perform coronary artery bypass grafting in a cohort of patients was determined by each prioritisation strategy, and mean lifetime costs and quality adjusted life years (QALYs) were compared.

DATA SOURCES Swedish Coronary Angiography and Angioplasty Registry (9935 patients with stable angina awaiting coronary artery bypass grafting and then followed up for cardiovascular events after the procedure for 3.8 years), and meta-analyses of prognostic effects (relative risks) of biomarkers.

RESULTS The observed risk of cardiovascular events while on the waiting list for coronary artery bypass grafting was 3 per 10,000 patients per day within the first 90 days (184 events in 9935 patients).

Using a cost effectiveness threshold of pound20,000- pound30,000 (euro22,000-euro33,000; \$32,000-\$48,000) per additional QALY, a prioritisation strategy using a risk score with estimated glomerular filtration rate was the most cost effective strategy (cost per additional QALY was < pound410 compared with the Ontario urgency score).

The impact on population health of implementing this strategy was 800 QALYs per 100,000 patients at an additional cost of pound 245,000 to the National Health Service.

The prioritisation strategy using a risk score with C reactive protein was associated with lower QALYs and higher costs compared with a risk score using estimated glomerular filtration rate.

CONCLUSION Evaluating the cost effectiveness of prognostic biomarkers is important even when effects at an individual level are small.

Formal prioritisation of patients awaiting coronary artery bypass grafting using a routinely assessed biomarker (estimated glomerular filtration rate) along with simple, routinely collected clinical information was cost effective.

Prioritisation strategies based on the prognostic information conferred by C reactive protein, which is not currently measured in this context, or a combination of C reactive protein and estimated glomerular filtration rate, is unlikely to be cost effective.

The widespread practice of using only implicit or informal means of clinically ordering the waiting list may be harmful and should be replaced with formal prioritisation approaches.

Document metadata

Corpus | SciFact

Document ID | 13625993

Evidence checklist

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9. **CONCLUSION** Evaluating the cost effectiveness of prognostic biomarkers is important even when effects at an individual level are small.
10. Formal prioritisation of patients awaiting coronary artery bypass grafting using a routinely assessed biomarker (estimated glomerular filtration rate) along with simple, routinely collected clinical information was cost effective.
11. Prioritisation strategies based on the prognostic information conferred by C reactive protein, which is not currently measured in this context, or a combination of C reactive protein and estimated glomerular filtration rate, is unlikely to be cost effective.
12. The widespread practice of using only implicit or informal means of clinically ordering the waiting list may be harmful and should be replaced with formal prioritisation approaches.

Source continuity

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